

Belief and skill in a navigation agent: a mechanistic-interpretability comparison of PPO+GRU and DreamerV3

Crusoe–Cogniland bridge_tunnel project

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Abstract

We study, in a partially observable grid-navigation task, how a model-free (PPO+GRU) and a model-based (DreamerV3) agent represent two latent factors: *belief* (which map type they are on) and *skill* (which obstacle-crossing tool they commit to). Using a controlled environment that supplies ground-truth belief and skill labels at every timestep, we (i) probe both factors from the agents’ recurrent states on held-out maps, (ii) test whether they are representationally separable, and (iii) intervene causally by steering. We find that the DreamerV3 RSSM deterministic state is a markedly cleaner belief carrier than the PPO GRU state (linear belief decodability 0.64 vs 0.38; ordinal Spearman 0.67 vs 0.07), while the GRU is the better skill carrier (0.60 vs 0.49). The apparent belief–skill entanglement is shown to be a *label confound* (it vanishes at fixed belief) and a belief-free skill code provably exists. Both factors are causal levers—skill in PPO, belief in Dreamer—but only when injected at the *read-out* (actor input), not the recurrent carry. We frame all claims against the POMDP sufficiency theorem and an explicit ladder of evidence (information $\rightarrow$ decodability $\rightarrow$ causal use).

1 Introduction

A competent agent in a partially observable environment must, by the classical sufficiency theorem (Åström 1965), maintain a sufficient statistic of its history—a *belief*. We ask: for two agent families that solve the same task, (a) is belief about a task-relevant latent decodable from the recurrent state, (b) is it separable from the agent’s committed *skill*, and (c) is either factor a causal lever on behaviour? We instrument a navigation task that makes belief and skill observable, and compare a model-free PPO+GRU agent to a model-based DreamerV3 agent.

2 Background: the POMDP formalism

A POMDP is a tuple $\langle \mathcal{S}, \mathcal{A}, p, R, \Omega, O, \gamma \rangle$ with hidden state $\mathcal{S}$, actions $\mathcal{A}$, observations Ω , transition $p(s' | s, a)$, reward $R(s, a)$, observation model $O(o | s', a)$, and discount γ . The agent never sees s_t ; it receives $o_{t+1} \sim O(\cdot | s_{t+1}, a_t)$ and its only access to the world is the history

$$\xi_t = (o_1, a_1, o_2, \dots, a_{t-1}, o_t) \in \Xi_t, \quad \pi : \Xi_t \rightarrow \Delta(\mathcal{A}). \quad (1)$$

Belief. The belief state is the posterior over the hidden state,

$$b_t(s) = \Pr(S_t = s | \xi_t), \quad b_t \in \Delta(\mathcal{S}) = \left\{ b \in \mathbb{R}^{|\mathcal{S}|} : b(s) \geq 0, \sum_s b(s) = 1 \right\}. \quad (2)$$

Belief update (Bayes filter). It evolves recursively via predict then correct,

$$\bar{b}_{t+1}(s') = \sum_s p(s' | s, a_t) b_t(s), \quad (3)$$

$$b_{t+1}(s') = \frac{O(o_{t+1} | s', a_t) \bar{b}_{t+1}(s')}{\sum_x O(o_{t+1} | x, a_t) \bar{b}_{t+1}(x)} \equiv [\tau(b_t, a_t, o_{t+1})](s'). \quad (4)$$

The belief is Markov: b_{t+1} depends on the past only through b_t .

Sufficiency (Åström 1965). b_t is a sufficient statistic of ξ_t for optimal control: there exists $\pi^*(b_t)$, and the POMDP reduces to a belief-MDP over $\Delta(\mathcal{S})$ with $\bar{R}(b, a) = \sum_s b(s) R(s, a)$ and transition induced by τ . Its optimal value is piecewise-linear and convex in b (Smallwood & Sondik 1973).

Why probe the recurrent state. A recurrent policy has the same functional form as the filter, $h_t = f_\theta(h_{t-1}, a_{t-1}, o_t) \approx \tau(b_{t-1}, a_{t-1}, o_t)$, so h_t *encodes* ξ_t and approximates b_t . Sufficiency *entails* that control-relevant state factors are recoverable from h_t as some function; it does *not* entail they are linearly decodable, separable, or causally used. Those are the empirical questions below, and we keep an explicit ladder: **(1) information present** (theorem), **(2) linearly decodable** (probing; correlational, Hewitt & Liang 2019), **(3) causally used** (steering; the strong claim).

Skill as an option. A skill is formalised as an option $\omega = \langle \mathcal{I}_\omega, \pi_\omega, \beta_\omega \rangle$ (Sutton, Precup & Singh 1999): initiation set, intra-option policy, termination. In our task the skill is a build/mine commitment with a one-shot, irreversible β .

3 Setup

Environment. A POMDP grid task: the agent sees a 21×21 egocentric crop and must reach a target. Obstacles can be crossed by converting them (*build*: water→wood; *mine*: rock→grass) or avoided. **Two variants:** BT (uncontrolled—tools always available, no labels) and BTC (controlled—3 labelled map categories {balanced, lakes, rocky} = ground-truth belief; an irreversible build/mine commitment = ground-truth skill). Crossed with two agents (PPO+GRU, DreamerV3) this gives the 2×2 design (Fig. 1).

4 Architectures

5 Datasets and methods

Four self-contained activation datasets (BT/BTC $\times$ PPO/Dreamer; 0.32–0.67 M timesteps each) store per-step activation sources, the full observation, and belief/skill/segment labels computed from map geometry. Probes are evaluated on **held-out maps** (grouped split) so accuracy is the honest generalisable decodability. We fit (i) a 3-class logistic probe for belief and skill, (ii) a ridge probe on the ordinal water-rock belief axis, (iii) difference-of-means directions $d_{\text{belief}} = \mu_{\text{lakes}} - \mu_{\text{rocky}}$, $d_{\text{skill}} = \mu_{\text{build}} - \mu_{\text{mine}}$, and measure entanglement by cosine, subspace principal angles, and projection fractions.

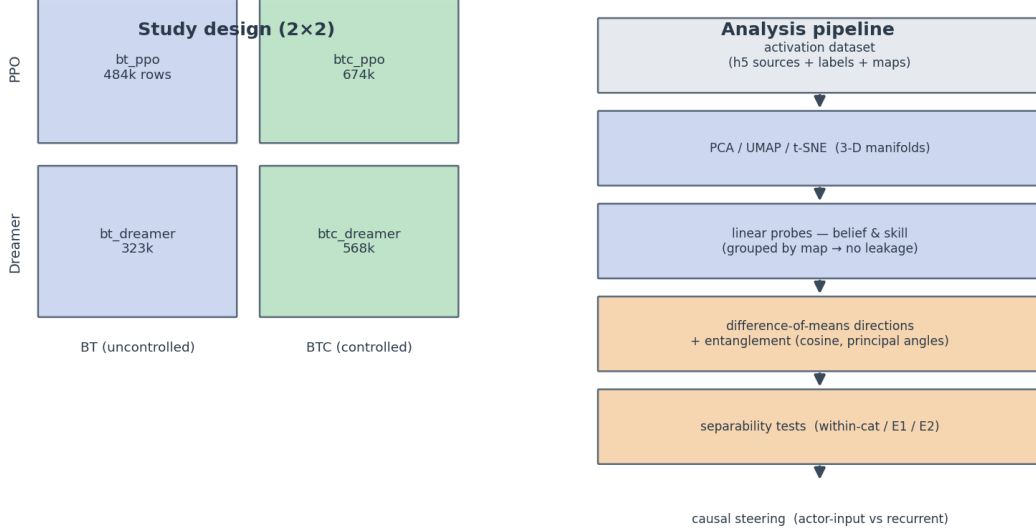


Figure 1: The 2×2 design and analysis pipeline. Four activation datasets on the same held-out maps; each run through DR, map-grouped probes, difference-of-means directions, separability tests, and causal steering.

6 Results

6.1 Geometry of the representations

Table 1 reports held-out decodability (chance 0.33).

source	belief acc	ordinal R^2 / ρ	skill acc	$\cos(b, s)$
PPO <code>gru_h</code>	0.38	-0.33 / $+0.07$	0.60	$+0.59$
PPO <code>enc_embed</code>	0.43	$+0.10$ / $+0.38$	0.53	$+0.67$
Dreamer <code>rssm_deter</code>	0.64	$+0.38$ / $+0.67$	0.49	$+0.53$
Dreamer <code>enc_embed</code>	0.59	$+0.47$ / $+0.72$	0.51	$+0.84$
Dreamer <code>stoch-logits</code>	0.51	$+0.19$ / $+0.48$	0.44	$+0.71$

Table 1: The model-based RSSM `rssm_deter` is a far cleaner belief carrier than the model-free `gru_h`; the GRU is the better skill carrier.

6.2 Belief and skill are separable (the entanglement is a confound)

	cos global	within balanced	E2 full off-type	E2 belief-removed off-type
PPO <code>gru_h</code>	$+0.58$	$+0.00$	0.37	0.62
Dreamer <code>rssm_deter</code>	$+0.54$	$+0.12$	0.33	0.83

Table 2: The global belief–skill cosine collapses at fixed belief (within-category control). A build-vs-mine probe is below chance on against-type commits with full features (it is a belief detector) but generalises once belief is projected out (E2)—a genuine belief-free skill code.

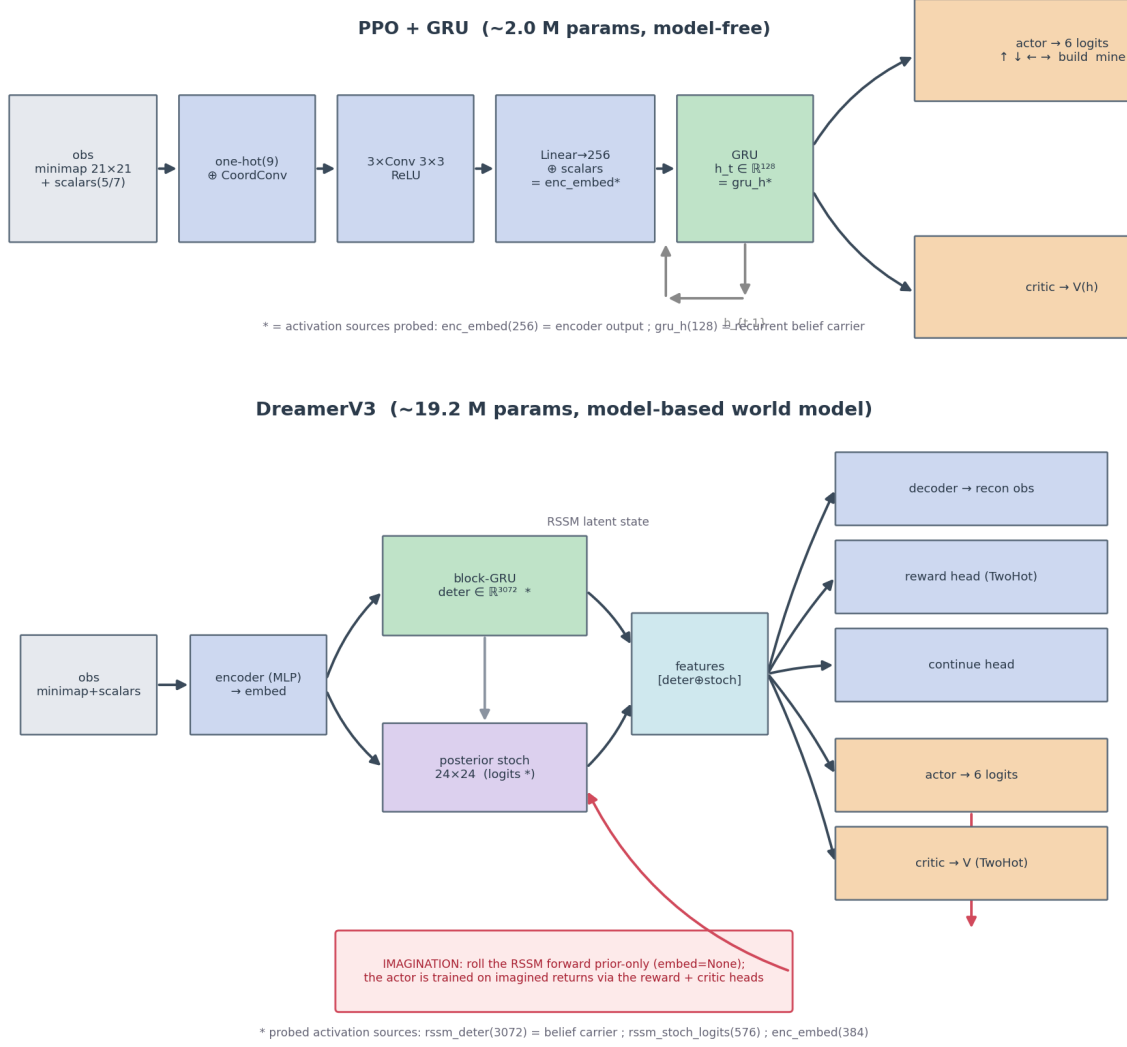


Figure 2: PPO+GRU (~2.0M params): the recurrent `gru_h` is the belief carrier. DreamerV3 (~19.2M): the RSSM `rssm_deter` is the belief carrier; the policy is trained in imagination via the reward head, not the decoder.

6.3 Causal steering

Injecting at the *actor input* (read-only) steers behaviour while preserving success; persistent injection into the recurrent carry breaks the agent (Fig. 6). For PPO, the skill commitment is controllable on balanced maps ($P(\text{mine}) = 0.70$ at $\alpha = -6$, $P(\text{build}) = 0.50$ at $\alpha = +6$, reach 0.8–1.0). For Dreamer, the belief axis monotonically shifts the commit with reach 0.85–0.98 (with a sign caveat from the confounded difference-of-means axis).

6.4 Behaviour and imagination

The trained Dreamer reproduces the PPO-like map→skill commit matrix only after exploration was raised (Fig. 7). Under a fair stochastic eval PPO beats Dreamer by 3–6 points with $\sim 10\times$ fewer parameters; argmax hurts *both* (the stochastic policy is the better controller). Finally, the world model does *not* faithfully imagine the rare build/mine tile conversions (bridge 0/4, tunnel



Figure 3: Belief×skill plane (PPO `gru_h` left, Dreamer `rssm_deter` right): the same cloud coloured by category and by skill. Both look diagonally entangled ($\cos \approx 0.5\text{--}0.6$); §6.2 shows this is a label confound.

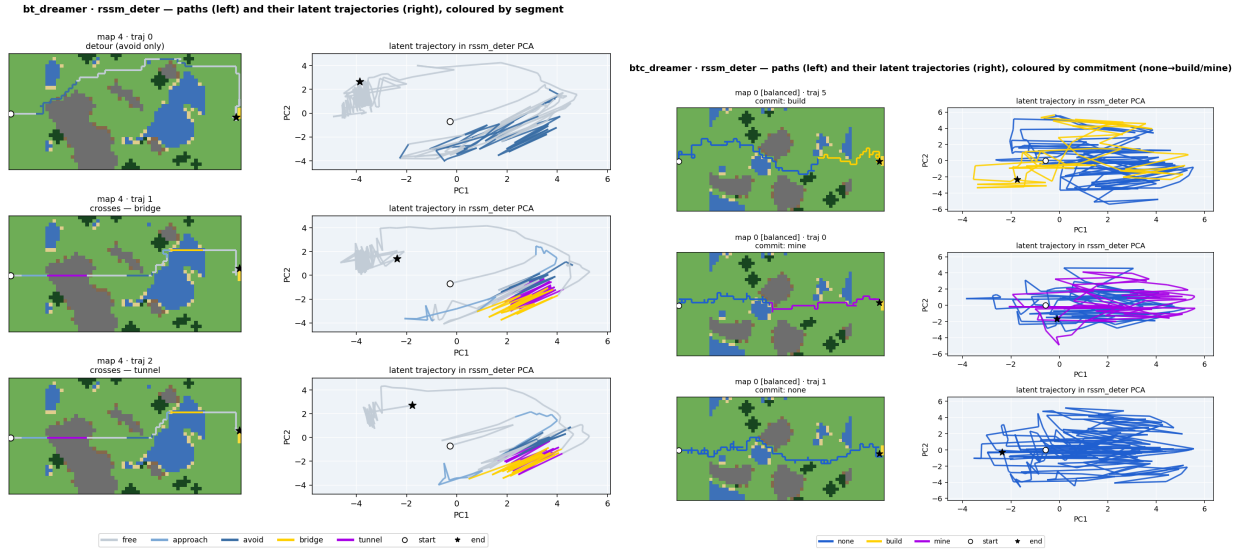


Figure 4: Paired map↔latent trajectories (Dreamer). Left (BT, seed 10004): detour/bridge/tunnel coloured by segment. Right (BTC, map 0): none/build/mine, blue until the commit point then yellow/purple. The route is visible simultaneously on the map and as a colour change in the latent path.

1/4 in a forced counterfactual) and long open-loop dreams drift—yet behaviour is correct, because the actor is trained from the reward head over latents, not from the decoder.

7 Discussion

Åström sufficiency guarantees the *information* (belief, and the commit state, which is itself a state variable) is recoverable from h_t , but not its representational form. Our probes establish linear decodability and our steering establishes causal use; only with the latter do we claim h_t “encodes/uses” a factor. Two limitations: (i) we study the task-relevant *marginal* of the belief (map type / water fraction), not the full posterior; (ii) the parameter gap ($\sim 9.5\times$) means Dreamer’s cleaner belief may partly reflect capacity, not only the model-based objective.

References

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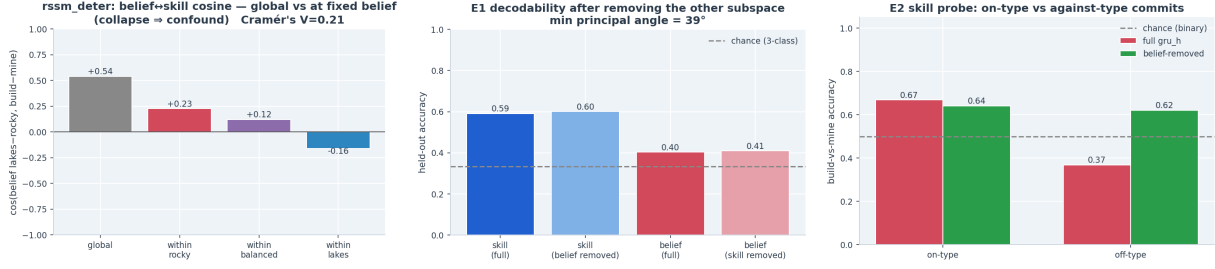


Figure 5: Separability tests (Dreamer `rssm_deter` / PPO). Within-category control; E1 subspace removal (each factor survives removing the other; min angle ~ 36 – 39°); E2 off-type generalisation.

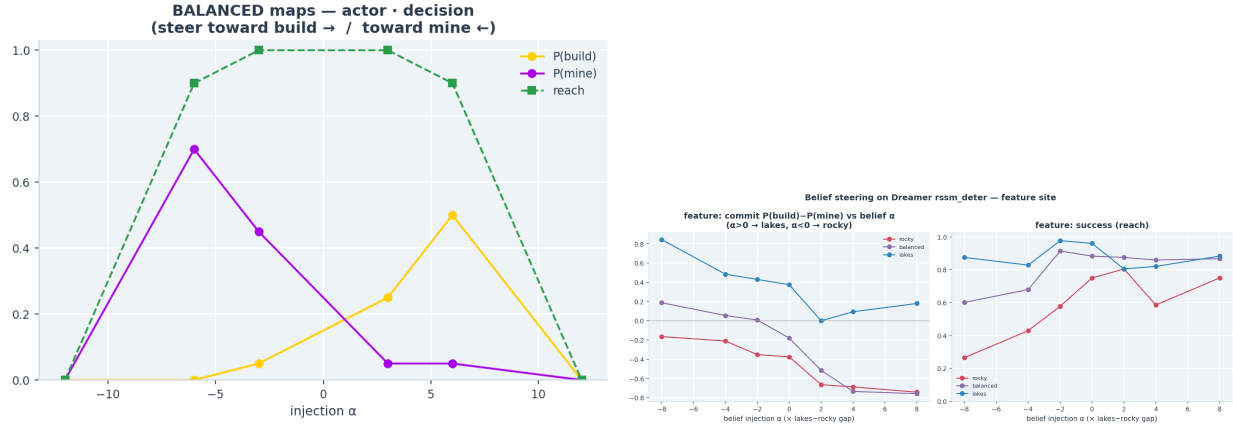


Figure 6: Left: PPO skill steering (actor-input, decision direction)—monotonic commit control with preserved reach. Right: Dreamer belief steering (feature site)—belief axis moves the committed skill across categories with success intact.

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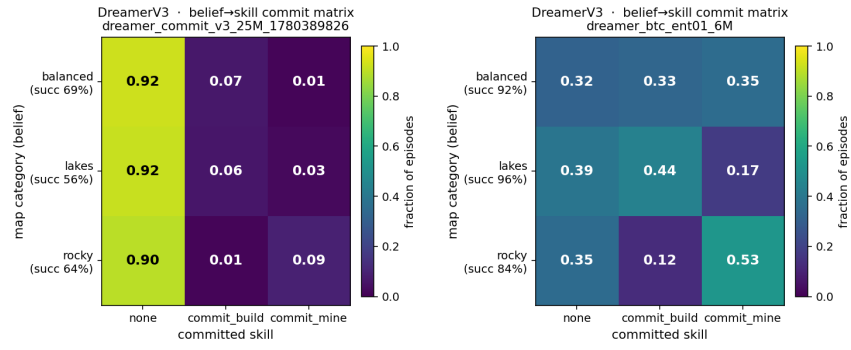


Figure 7: Dreamer BTC commit matrix: baseline (~ 0.92 *none* everywhere) vs after raising exploration (lakes $\rightarrow$ build 0.44, rocky $\rightarrow$ mine 0.53).